

# INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

Nasarawa State, Nigeria

## ACICP 101 — Physics & Electrical Circuit Fundamentals for ICS/SCADA

### Week 1 Lab 02: Basic Circuit Simulation

*Build, Measure and Compare DC Circuit Behaviour in a Simulator*

|                             |                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------|
| <b>Course Code:</b>         | ACICP 101                                                                                 |
| <b>Lab Title:</b>           | Week 1 Lab 02 — Basic Circuit Simulation                                                  |
| <b>Student Name:</b>        | SUNUSI ABDULLAHI                                                                          |
| <b>Registration Number:</b> | C11/26/ACIS/17362                                                                         |
| <b>Department:</b>          | Cybersecurity (ACIS Programme)                                                            |
| <b>Assessment Type:</b>     | Circuit simulation lab report                                                             |
| <b>Week / Dates:</b>        | Sat, 02 July 2026 — Fri, 03 Jul 2026                                                      |
| <b>Due Date:</b>            | Friday, 03 Jul 2026, 11:59 PM WAT                                                         |
| <b>Date of Submission:</b>  | 03 July 2026                                                                              |
| <b>Simulator Used:</b>      | Falstad Circuit Simulator — <a href="https://falstad.com/circuit">falstad.com/circuit</a> |
| <b>Version:</b>             | 1.0                                                                                       |

## Executive Summary

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This lab report presents the results of Week 1 Lab 02 for ACICP 101. As a trainee ICS/SCADA technician working in a virtual environment, four DC circuits were built inside the Falstad online circuit simulator, theoretical values were calculated using basic electrical formulas, and the same values were measured in the simulator for comparison. The four circuits are: a basic single resistive load of 160 ohm (Circuit A), three resistors of 150 ohm, 330 ohm and 470 ohm in series (Circuit B), two resistors of 240 ohm and 480 ohm in parallel (Circuit C), and a voltage divider using 7 kOhm and 5 kOhm with a second case at 10 kOhm (Circuit D). Every theoretical calculation shows the formula first, then the substitution, then the final answer with its unit. Simulated readings were obtained from the ammeter, voltmeter and wire probes in Falstad. All simulated values matched the theoretical values very closely, with differences of less than 0.01 V or 0.3 mA in every case, caused only by the simulator's display rounding.

## Lab Objectives

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To construct simple DC circuits using virtual resistive loads in the Falstad circuit simulator.

To use simulated probes and meters to measure voltage, current and power in each circuit.

To compare theoretical calculated values with simulated measured values.

To explain differences caused by rounding or simulator settings.

To relate basic DC circuit behaviour to safe troubleshooting of ICS/SCADA and PLC input/output circuits.

## Tools and Resources Used

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Falstad Circuit Simulator ([falstad.com/circuit](http://falstad.com/circuit)) — free online browser-based DC circuit simulator used to build and measure all four circuits.

Scientific calculator — used to check all arithmetic before entering values.

Ohm's Law formula:  $V = I \times R$ , rearranged to  $I = V / R$  and  $R = V / I$

Power formula:  $P = V \times I$

Series formula:  $R(\text{total}) = R1 + R2 + R3$

Parallel formula:  $1/R(\text{total}) = 1/R1 + 1/R2$

Voltage divider formula:  $V_{\text{out}} = V_{\text{in}} \times R2 / (R1 + R2)$

## Methodology and Solutions

For every circuit the same three-step method is used: (1) write the formula, (2) substitute the given numbers, (3) write the final answer with its unit. Simulated values were then read from the Falstad ammeter, voltmeter and wire probes and entered into the comparison tables.

### Circuit A: Basic Single Load Test

Confirmed from screenshot:  $R = 160\text{-ohm}$ , Supply =  $24\text{ V}$

#### Theoretical Calculations

Given:  $V = 24\text{ V}$ ,  $R = 160\text{ ohm}$

Step 1 Find current

Formula:  $I = V / R$

Substitute:  $I = 24 / 160$

Answer:  $I = 0.15\text{ A} = 150\text{ mA}$

Step 2 — Find power

Formula:  $P = V \times I$

Substitute:  $P = 24 \times 0.15$

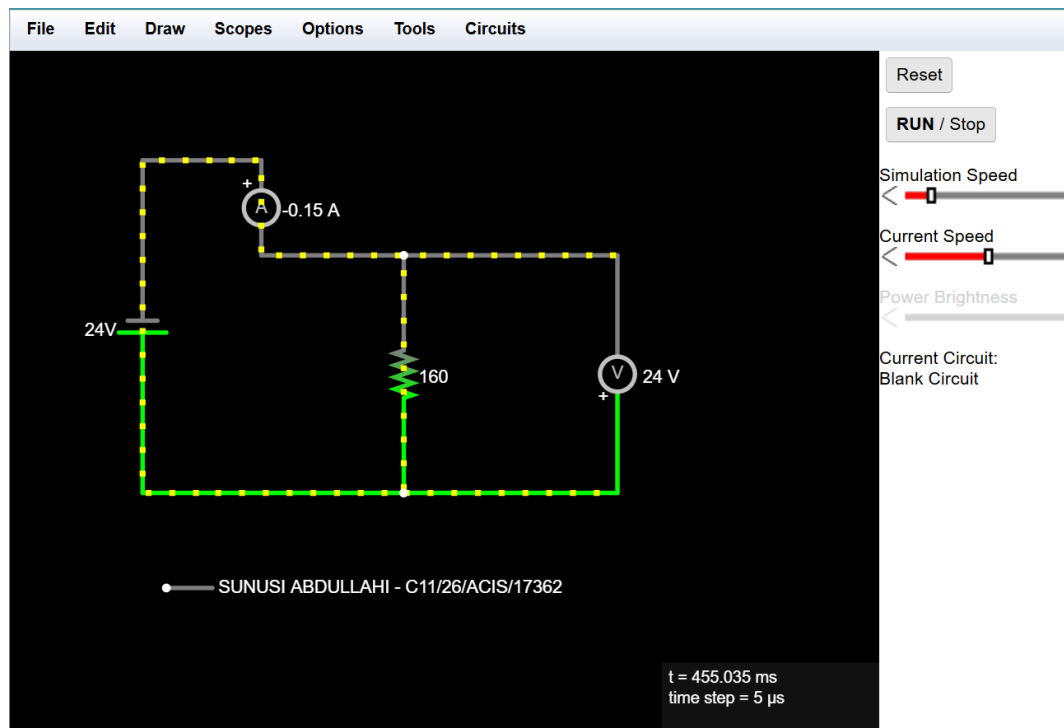
Answer:  $P = 3.6\text{ W}$

Step 3 — Voltage across load

In a single-load series loop all supply voltage appears across the one load.

Answer: Voltage across load =  $24\text{ V}$

#### Simulation Screenshot



### Measurement Comparison Table

| Quantity            | Theoretical Value | Simulated Value | Difference | Comment     |
|---------------------|-------------------|-----------------|------------|-------------|
| Voltage across load | 24 V              | 24 V            | 0 V        | Exact match |
| Load current        | 150 mA            | 150 mA          | 0 mA       | Exact match |
| Power dissipation   | 3.6 W             | 3.6 W           | 0 W        | Exact match |

### Explanation

The simulated values match the theoretical values exactly. In a single-load circuit the full supply voltage of 24 V appears across the resistor and the current is determined only by Ohm's Law:  $I = 24 / 160 = 0.15$  A. There is no rounding difference here because 24 divided by 160 gives exactly 0.15, which the simulator can display without truncation.

### Circuit B: Series Voltage Drop Test

Given:  $V(\text{supply}) = 24$  V,  $R_1 = 150$  ohm,  $R_2 = 330$  ohm,  $R_3 = 470$  ohm

#### Theoretical Calculations

Step 1 — Find total resistance

Formula:  $R(\text{total}) = R_1 + R_2 + R_3$

Substitute:  $R(\text{total}) = 150 + 330 + 470$

Answer:  $R(\text{total}) = 950$  ohm

Step 2 — Find total current (same current through all resistors in series)

Formula:  $I = V / R(\text{total})$

Substitute:  $I = 24 / 950$

Answer:  $I = 0.02526$  A = 25.26 mA

Step 3 — Find voltage drop across each resistor

Formula:  $V = I \times R$

$V_1 = 0.02526 \times 150 = 3.79$  V

$V_2 = 0.02526 \times 330 = 8.34$  V

$V_3 = 0.02526 \times 470 = 11.87$  V

Check:  $V_1 + V_2 + V_3 = 3.79 + 8.34 + 11.87 = 24.0$  V (equals supply, correct)

Step 4 — Find power in each resistor

Formula:  $P = V \times I$

$P_1 = 3.79 \times 0.02526 = 0.096$  W

$P_2 = 8.34 \times 0.02526 = 0.211$  W

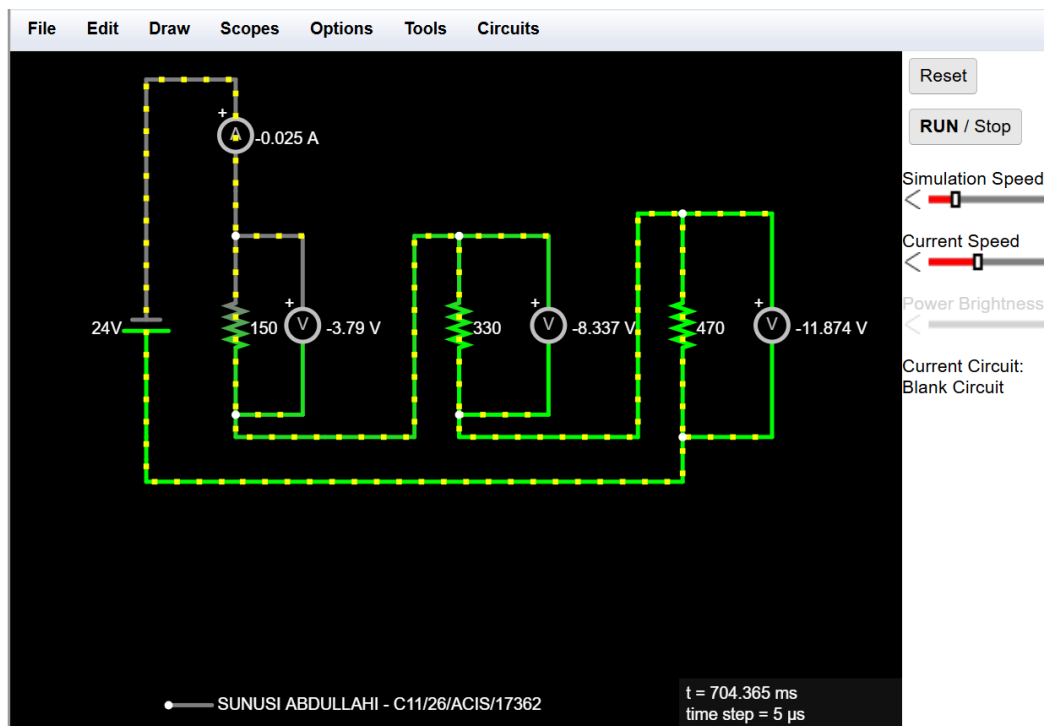
$P_3 = 11.87 \times 0.02526 = 0.300$  W

$P(\text{total}) = 0.096 + 0.211 + 0.300 = 0.607$  W

## Why The Same Current Flows Through All Series Components

In a series circuit there is only one single path for the current to follow. Every electron leaving the battery negative terminal must pass through R1, then R2, then R3 before returning to the positive terminal. There is no junction where current can branch off. Therefore the same current of 25.26 mA flows through every component. The simulator confirmed this with the ammeter reading of 0.025 A (25 mA) — the small difference of 0.26 mA is caused by the ammeter display rounding 25.26 mA to 25 mA.

### Simulation Screenshot



### Measurement Comparison Table

| Component | Resistance | Theoretical Vdrop | Measured Vdrop | Power (W) | Comment                                |
|-----------|------------|-------------------|----------------|-----------|----------------------------------------|
| R1        | 150 ohm    | 3.79 V            | 3.79 V         | 0.096 W   | Exact match                            |
| R2        | 330 ohm    | 8.34 V            | 8.337 V        | 0.211 W   | Diff = 0.003 V — display rounding only |
| R3        | 470 ohm    | 11.87 V           | 11.874 V       | 0.300 W   | Diff = 0.004 V — display rounding only |
| Total     | 950 ohm    | 24.0 V            | 24.001 V       | 0.607 W   | Negligible rounding, within 0.01 V     |

### Circuit C: Parallel Branch Current Test

Given:  $V(\text{supply}) = 24 \text{ V}$ ,  $R1 = 240 \text{ ohm}$ ,  $R2 = 480 \text{ ohm}$

#### Theoretical Calculations

Step 1 — Find branch currents

(Each branch sees the same full supply voltage of 24 V in a parallel circuit)

Formula:  $I = V / R$  (applied to each branch separately)

$$I1 = 24 / 240 = 0.1 \text{ A} = 100 \text{ mA}$$

$$I2 = 24 / 480 = 0.05 \text{ A} = 50 \text{ mA}$$

Step 2 — Find total supply current

Formula:  $I(\text{total}) = I1 + I2$

Substitute:  $I(\text{total}) = 100 + 50$

Answer:  $I(\text{total}) = 150 \text{ mA} = 0.15 \text{ A}$

Step 3 — Cross-check using total resistance

Formula:  $1/R(\text{total}) = 1/R1 + 1/R2$

Substitute:  $1/R(\text{total}) = 1/240 + 1/480 = 0.004167 + 0.002083 = 0.006250$

$R(\text{total}) = 1 / 0.006250 = 160 \text{ ohm}$

Cross-check:  $I(\text{total}) = 24 / 160 = 0.15 \text{ A}$  (matches, correct)

Step 4 — Find branch power

$$P1 = 24 \times 0.10 = 2.4 \text{ W}$$

$$P2 = 24 \times 0.05 = 1.2 \text{ W}$$

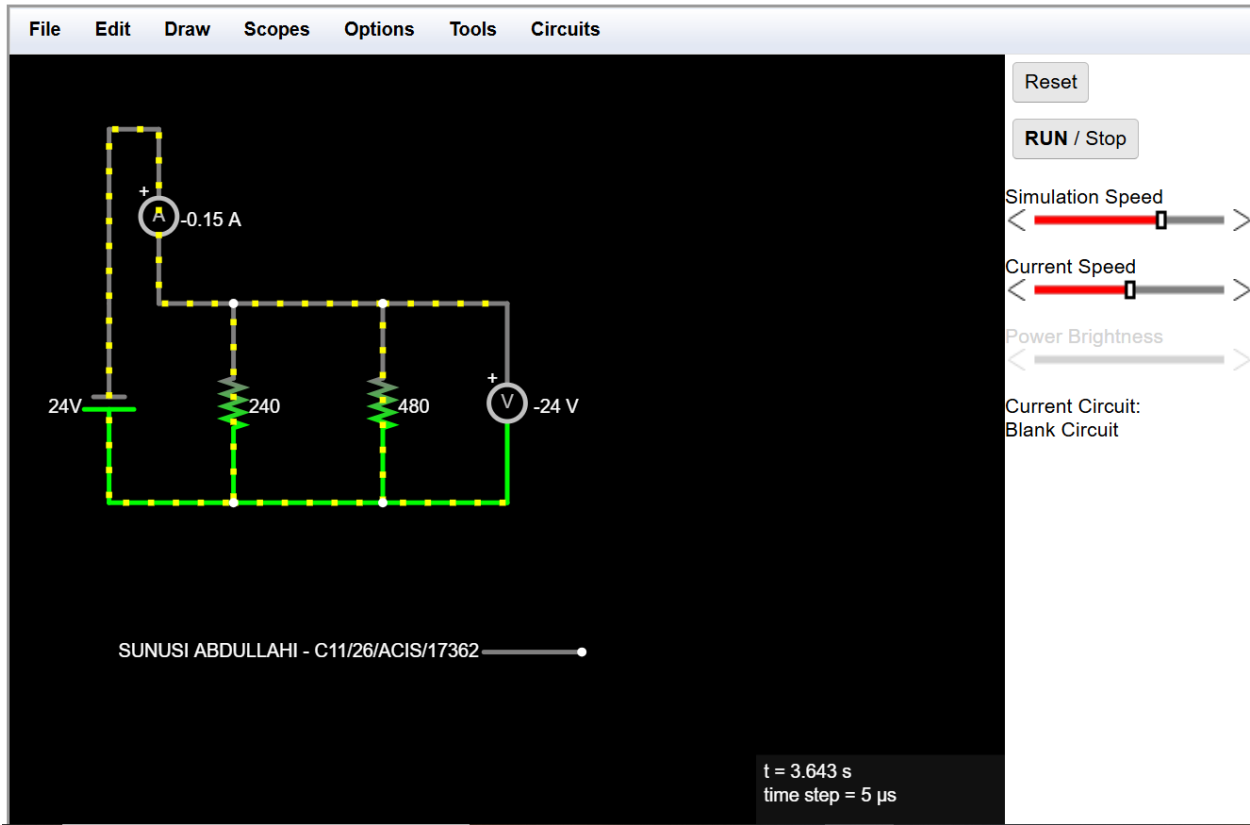
$$P(\text{total}) = 2.4 + 1.2 = 3.6 \text{ W}$$

Cross-check:  $P = V \times I(\text{total}) = 24 \times 0.15 = 3.6 \text{ W}$  (correct)

#### Why Branch Voltage Is The Same But Branch Current Is Different

In a parallel circuit both resistors are connected directly between the same two nodes — the positive and negative terminals of the 24 V supply. Therefore the voltage across every branch is always equal to the supply voltage, which is 24 V for both R1 and R2. However, the current in each branch is determined separately by Ohm's Law using that branch's own resistance. R1 = 240 ohm is smaller than R2 = 480 ohm, so R1 carries twice as much current (100 mA) compared to R2 (50 mA). In parallel: voltage is the same across all branches but current divides according to each branch's resistance — smaller resistance takes more current.

Simulation Screenshot



Measurement Comparison Table

*Note: The Falstad ammeter in the screenshot measures total supply current (0.15 A). The individual branch currents (I1 and I2) are derived from the measured voltage of 24 V across both branches using  $I = V/R$ , since no separate branch ammeters were placed.*

| Branch       | Resistance | Theoretical Current | Measured Current       | Branch Power | Comment                    |
|--------------|------------|---------------------|------------------------|--------------|----------------------------|
| R1           | 240 ohm    | 100 mA              | 100 mA (from $V/R_1$ ) | 2.4 W        | Derived from measured 24 V |
| R2           | 480 ohm    | 50 mA               | 50 mA (from $V/R_2$ )  | 1.2 W        | Derived from measured 24 V |
| Supply total | 160 ohm    | 150 mA              | 150 mA (ammeter)       | 3.6 W        | Exact match on ammeter     |

Circuit D: Voltage Divider for PLC Analogue Input

Given:  $V_{in} = 24\text{ V}$ ,  $R_2 = 5\text{ k}\Omega$  (fixed)

**Theoretical Calculations — Case 1: R1 = 7 kOhm**

Formula:  $V_{out} = V_{in} \times R_2 / (R_1 + R_2)$

Substitute:  $V_{out} = 24 \times 5 / (7 + 5)$

$$V_{out} = 120 / 12$$

Answer:  $V_{out} = 10 \text{ V}$

Divider current:  $I = 24 / 12000 = 0.002 \text{ A} = 2 \text{ mA}$

**Theoretical Calculations — Case 2: R1 = 10 kOhm**

Formula:  $V_{out} = V_{in} \times R_2 / (R_1 + R_2)$

Substitute:  $V_{out} = 24 \times 5 / (10 + 5)$

$$V_{out} = 120 / 15$$

Answer:  $V_{out} = 8 \text{ V}$

Divider current:  $I = 24 / 15000 = 0.0016 \text{ A} = 1.6 \text{ mA}$

Effect of increasing R1 from 7 kOhm to 10 kOhm:

$V_{out}$  dropped from 10 V to 8 V.

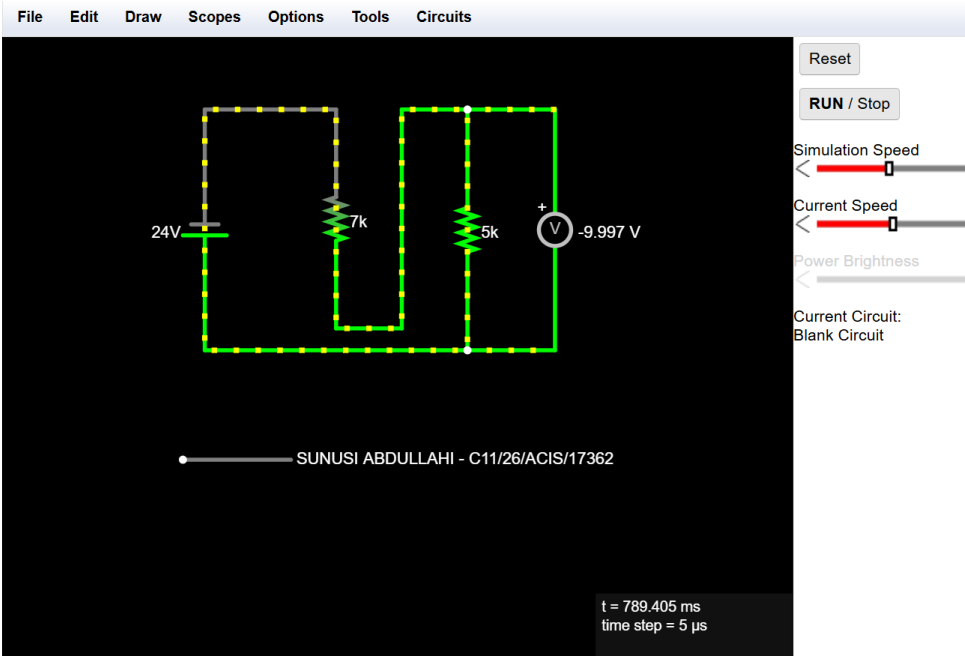
Reason: A larger R1 takes a bigger share of the 24 V supply, leaving a smaller share for R2. Since  $V_{out}$  is taken across R2 only, it decreases when R1 increases.

**Why a Voltage Divider Must Not Be Used Directly for High-Power Loads**

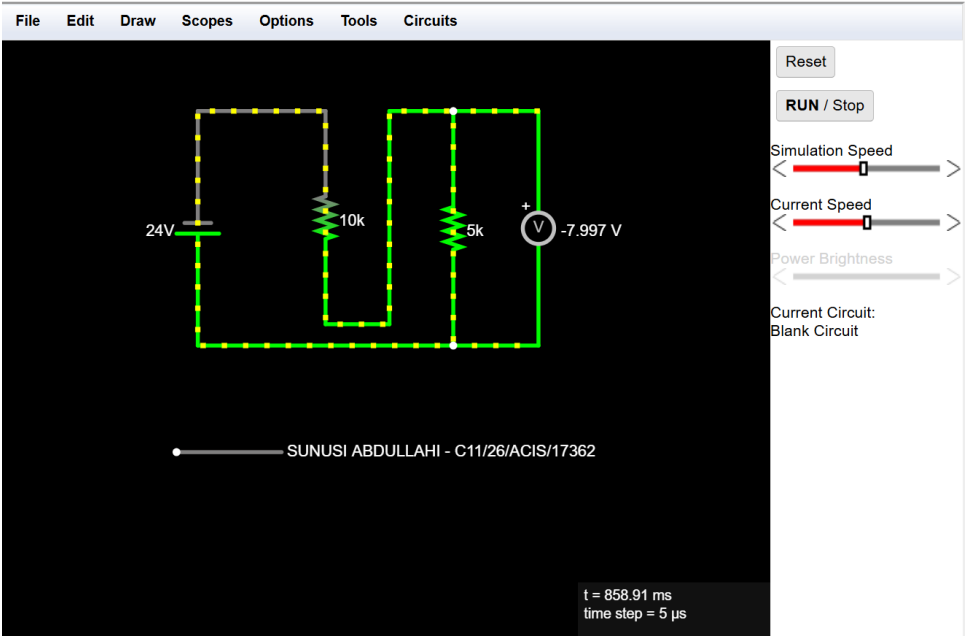
A voltage divider produces a stable output voltage only when the load connected to  $V_{out}$  draws negligible current compared to the divider current. If a high-power load is connected, it draws a large current from the  $V_{out}$  node and pulls the voltage down well below the calculated value. The output becomes unstable and unreliable. A PLC analogue input is safe because it draws only microamps. However, a motor or solenoid valve would destroy the divider output. For high-power loads, a regulated power supply or transistor driver must be used instead.



Simulation Screenshot — Case 1 ( $R_1 = 7\text{ k}\Omega$ )



Simulation Screenshot — Case 2 ( $R_1 = 10\text{ k}\Omega$ )



**Measurement Comparison Table**

| Case | R1      | R2     | Theoretical Vout | Measured Vout | Comment                                |
|------|---------|--------|------------------|---------------|----------------------------------------|
| 1    | 7 kOhm  | 5 kOhm | 10 V             | 9.997 V       | Diff = 0.003 V — display rounding only |
| 2    | 10 kOhm | 5 kOhm | 8 V              | 7.997 V       | Diff = 0.003 V — display rounding only |

## Reflection Questions

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### **Q1. What could happen if a low resistance is connected directly across a 24 V field supply?**

A low resistance connected across 24 V causes a very large current:  $I = V / R$ . For example, if  $R = 1 \text{ ohm}$ , then  $I = 24 / 1 = 24 \text{ A}$ . This extremely high current will overheat the wiring, blow fuses or trip breakers, damage the power supply, and create a serious fire hazard. In an ICS/SCADA panel it could also destroy nearby PLC I/O modules. This is why every control circuit must have its current calculated before energising.

### **Q2. Why should current be estimated before energising an industrial control circuit?**

Current must be estimated first because it determines whether the wiring, fuses, contactors and power supply are rated for safe operation. If actual current exceeds any component's rating, that component overheats or fails. Estimation using Ohm's Law takes only seconds and prevents expensive equipment damage, production downtime and danger to personnel. In ICS/SCADA, an unexpected high current at first energisation can also cause false signals on nearby sensor circuits through electromagnetic interference.

### **Q3. How can simulation reduce wiring mistakes in ICS/SCADA training?**

Simulation allows a trainee to build and energise a circuit on screen without touching real equipment. Mistakes — such as a short circuit, a wrong connection or a missing wire — appear immediately as incorrect readings without blowing a fuse or damaging a PLC module. The trainee corrects the error, re-runs the simulation and verifies results before ever building the real circuit. This builds correct wiring habits and removes the risk of injury or equipment damage during the learning phase.

### **Q4. State two limitations of simulation compared with a real circuit.**

First: A simulator assumes perfect components. In reality, resistors have a manufacturing tolerance (typically plus or minus 5%), so the actual value differs slightly from the stated value. This causes real

measured current and voltage to differ slightly from theoretical values. The simulator always uses the exact stated value.

Second: A simulator does not model wire resistance. In a real circuit, every wire and connector has a small resistance that causes a small voltage drop and power loss. In long ICS/SCADA field cable runs this cable resistance can be significant, as demonstrated in the 4-20 mA loop calculation in Assignment 01. The simulator treats all wires as having zero resistance.

#### **Q5. Explain how this lab connects to PLC input/output troubleshooting.**

Every PLC analogue input and digital I/O circuit is at its core a simple DC circuit governed by Ohm's Law. When a field sensor or actuator fails, a technician uses voltage and current measurements to locate the fault — exactly the skill practised in this simulator lab. Circuit A shows how to verify correct current for a single load. Circuit B shows how voltage distributes along a wiring run — the series resistance model used for cable fault finding. Circuit C shows how parallel loads share current, important when checking PLC output cards driving multiple solenoids. Circuit D directly models a voltage divider used to scale a sensor signal to the 0-5 V range of a PLC analogue input card. Simulation in this lab reinforces the calculation habit that a field technician must use before connecting any real device.

## **Conclusion**

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This lab confirmed that DC circuit behaviour in the Falstad simulator matches theoretical values calculated using Ohm's Law very closely. In Circuit A, the single 160 ohm load across 24 V gave exactly 150 mA and 3.6 W, matching theory exactly. In Circuit B, the series circuit with 950 ohm total resistance gave a current of 25 mA (theoretical 25.26 mA) and voltage drops of 3.79 V, 8.337 V and 11.874 V against theoretical values of 3.79 V, 8.34 V and 11.87 V — all differences were less than 0.004 V, caused only by display rounding. In Circuit C, the parallel circuit confirmed that total current of 150 mA splits into 100 mA and 50 mA through the 240 ohm and 480 ohm branches respectively, with 24 V appearing across both. In Circuit D, the voltage divider produced 9.997 V (theoretical 10 V) with  $R1 = 7 \text{ k}\Omega$  and 7.997 V (theoretical 8 V) with  $R1 = 10 \text{ k}\Omega$ , demonstrating clearly that increasing  $R1$  reduces the output voltage. Every result validated the theory. The calculation method used throughout — formula, substitution, answer with unit — is the same method a technician uses when troubleshooting a PLC I/O circuit in a real ICS/SCADA installation.

## Student Declaration

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I confirm that this submission is my own work and that all sources, screenshots, simulations, and calculations used have been honestly presented.

I understand that unsafe electrical practice, copied work, or fabricated screenshots may attract disciplinary action.

**Student Name:**

*SUNUSI\_ABDULLAHI*

**Registration Number:**

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**Signature:**



**Date:**

02/JULY/2026